

The Design and Engineering of an Accurate Bicycle Power Meter

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1. Introduction

A bicycle power meter is a device to measure the power input of a rider to the bicycle. It is used by athletes and coaches to monitor and improve performance using watts as the key metric.

This paper will describe the process of design and engineering of a bicycle power meter with drift free accuracy of the measured data being the defined goal.

1.1 Prior Art

Devices for measuring power input of a cyclist have been developed around most components in the drive train. There are examples of devices located in the pedals (Garmin Vector) [1], crank arms (Stages), crank spider (SRM, Quarq), chain (Polar) and rear wheel hub (Powertap,) [2]. The SRM and Powertap devices are considered the current market ‘gold standards’.

Very little scientific testing has been performed on any of these devices to establish their level of performance as accurate measurement instruments. One such study comparing SRM to Powertap showed that the devices demonstrated mean accuracies within a 50W to 1000W range consistent with the manufacturer claims of +/-2.5%. However temperature had a large effect on readings. In field trials the SRM average and max power readings were 5.2% and 8.4% lower than the Powertap device. [3] Both devices demonstrated data drift requiring regular recalibration.

On a practical level a coach training a rider using Powertap would be looking at different data and making different recommendations than for the same rider if he was using the SRM unit. Measurement drift means data from the same device could be misleading if not regularly calibrated.

There is a need for a power meter that offers consistent, accurate and drift free data.

2. Deriving Power From a Bicycle Crank

Power cannot be measured directly. Power is derived from torque multiplied by rotational speed, (angular velocity).

$$Power (P) = Torque (\tau) \times Angular\ velocity (\omega)$$

2.1 Measuring Cadence

Angular Velocity or, in cycling terms, Cadence, is simply the rate of rotation of the crank set, usually measured in revolutions per minute (RPM).

In order to use cadence in our power equation we must convert RPM to radians per second.

$$\omega = RPM * \frac{2\pi}{60}$$

Crank rotational speed can be measured simply with a switch that triggers every time the crank rotates. Counting how often that happens against a clock. A simple device for this is a reed switch; two small metallic and magnetised blades, (or reeds), sealed inside a tiny glass tube and held in a vacuum. Pass a magnet close by and the two reeds snap together closing a circuit and sending a count signal. As the magnet passes, the polarity of the blades reverses and they repel each other breaking the circuit. With that count against the clock angular velocity can easily, and precisely, be calculated.

2.2 Measuring torque.

There are a total of 6 loads [3] that are applied from the pedal to the crank, referring to figure 1 below;

1. Tangential load – This force generates power as the crank rotates. A power meter should accurately measure this force, and its equivalent torque (load times crank length) about the bottom bracket bearings.
2. Axial load – This force produces tension or compression in the crank arm. It is highest when the rider's weight acts downward on the pedal when the pedal is at its lowest point. This force produces no power. A power meter should have no output to this force.
3. Side load – This load can be present when the rider leans the bike during hard acceleration. This force produces no power. A power meter should have no output to this force.
4. Pedal axis torque – It is not possible to apply this load to the crank through the pedal unless the pedal bearings are faulty.
5. Crank axis torque (twisting of the crank) – This load is always present since the pedal centreline is necessarily offset from the crank. This torque produces no power. A power meter should have no output to this torque.
6. Side torque – This torque bends the crank towards or away from the frame. This load is highest when the rider's weight acts downward on the pedal when the pedal is at its lowest point. This torque produces no power. A power meter should have no output to this torque.

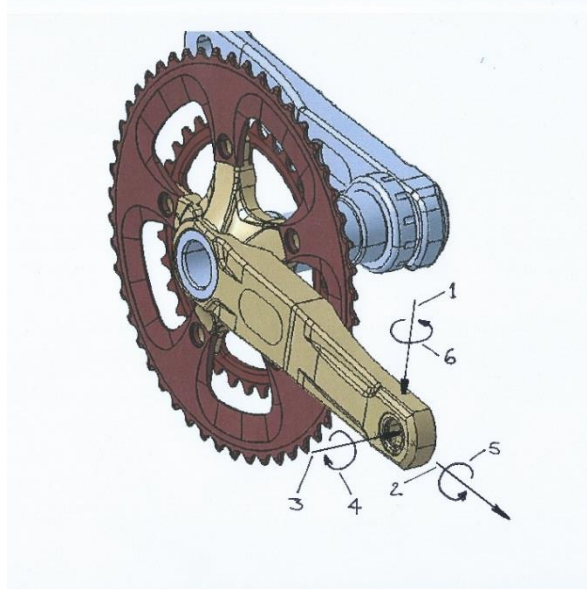


Figure 1. Combined loading of a bicycle crank

The problem is reduced to accurately measuring a single force or load, the tangential load, passing through the crank arm.

With reference to existing devices (1.1 above) it can be seen that the spider or pedal location for the measuring section are inherently complicated in this regard. Neither location allows for the simple isolation of the tangential load.

By measuring strains caused by the tangential load (the only load generating power), knowing the cross section of the material under strain, the stress the tangential load can be derived very precisely.

The principles of design of strain gauge force measurement devices (load cells), are well established, figure 2.

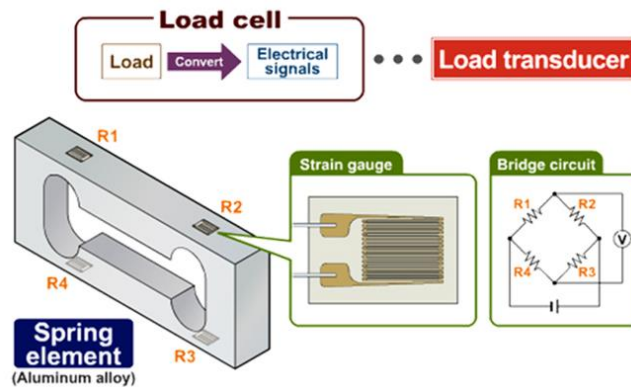


Figure 2. Typical Binocular Spring Beam Load Cell [4]

By designing a beam deliberately with a spring element that works only in the direction of interest strains can be controlled. Gauges can then be precisely placed to only measure the effective load of interest (1). The above device could be designed to be much stiffer in the plane of the page that any strain caused by a side load (3) would be negligible. The hole eliminates shear stresses and strains in the web, or side faces, of the beam. Similarly for twisting effects the beam can be designed to be so much stiffer that strains in these directions (6 & 5). The axial load (2) in this circumstance is trivial as all gauges would record a tension of the same value and are simply omitted from subsequent calculations.

Making the load cell massive in the unwanted load planes is impractical for a device to be used on a modern bicycle. Weight is a critical factor both for performance and as a marketing tool.

Unwanted load measurement can be eliminated by careful design and manufacture of the local section of the crank arm where the measurement will take place.

Figure 3 is an FE optimisation showing the typical load distribution through a generic NDS (non-drive side) crank arm. By placing the load cell in the relatively neutral zone immediately below the crank spindle, figure 4, and placing the gauges directly in the load path seen in each side wall, figure 5, a precision measuring device is fully integrated into the crank arm.

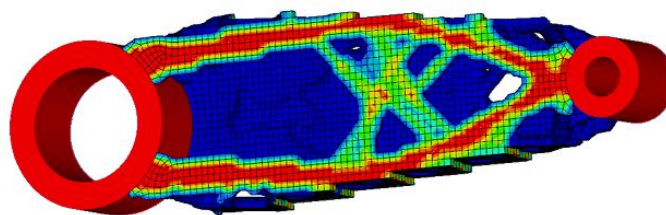


Figure 3. FE Optimisation of a Generic NDS crank arm [6]

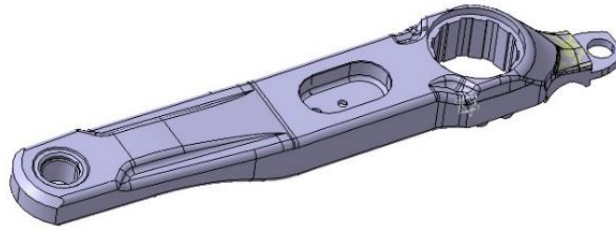


Figure 4. Crank arm with precision machined load cell integration

Once the geometry of the measurement section is defined the practical issue of strain gauging becomes critical. The surface of the gauge area is specially etched to achieve a perfect bond for the gauge. Instrument quality gauges, (generally thinner foils and backing structures) are placed to within less than a tenth of a millimetre under a microscope. The adhesive is oven cured for 5 hours. At the end of the process the measuring section of the gauge is an integral part of the material of the load cell section. Secondary effects of glue line thickness and stiffness that would attenuate the strain measurement are largely eliminated. The material of the gauge and the surface are matched for thermal effects allowing for precise temperature compensation in use.

Referring to figures 5 to 8 below it can be seen how the precise placement of the gauges and their cross body relationship in a full, four gauge, Wheatstone bridge eliminates all output other than the required tangential load.

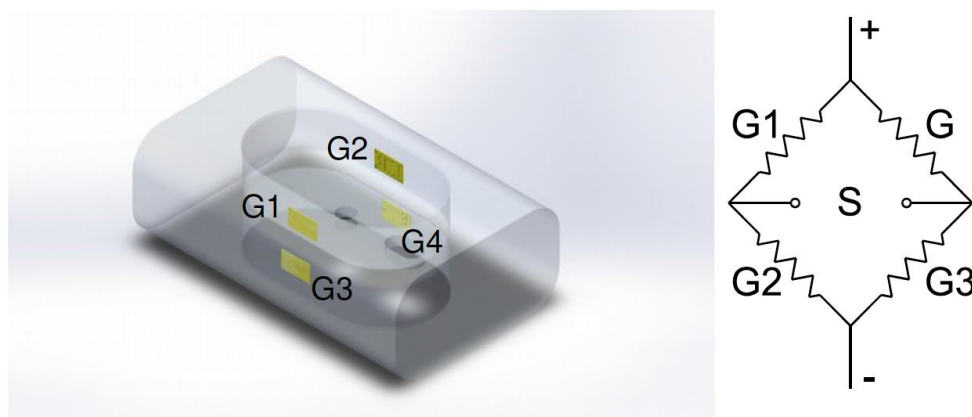
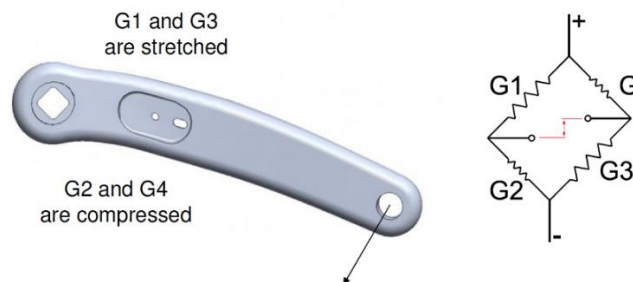


Figure 5. Strain gauge placement in load cell pocket [7]



Tangential Force = **Power Generated** = Unbalanced Bridge = Large Output

Figure 6. Operation of the crank arm integrated load cell to measure torque [8]

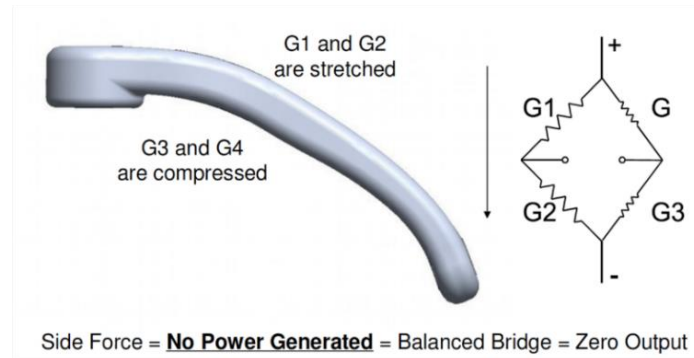


Figure 7. Zero output from lateral bending non-power generating load [9]



Figure 8. Zero output from twisting non-power generating load [10]

2.3 Power.

Measured load is combined with crank length to give torque. Torque data are recorded at a constant sampling rate of 256Hz.

A rotation is measured between consecutive cadence events. For best accuracy, the cadence magnet does not trigger during the power stroke

Average torque is calculated each crank rotation and transmitted via Ant+ wireless protocol to the display device.

Power updates are available every crank rotation.

3. Conclusion.

By applying known principles of load cell design to define a precision torque measuring device in the first instance then building a bicycle crank around that device an accurate, drift free power meter for a bicycle is established.

Independent certification of the device confirmed a maximum error of 0.11Nm for readings below 17Nm and 0.57% for measurements above 17Nm. [11] this exceeds typical industry claimed accuracies for existing devices of around +/-2%. [3, 12]

This data fidelity comes from:

- Precise cadence trigger using a reed switch.

- Symmetric gauge layout in a full Wheatstone bridge insensitive by design to non-power generating loads.
- Temperature compensated gauges in a full Wheatstone bridge, eliminates sensitivity to temperature effects.
- High sample rate
- High precision analog to digital conversion of strain gauge data with resolution of $0.3\mu\text{V}$ (0.025N.m torque)



Figure 9. Complete crank assembly prior to power meter installation.[13]

4 References

- [1] Ben Delaney Garmin Vector Long Term Review January 12 2014 www.bikeradar.com.
- [2] Kraig Willet 2011 Power Meter Review. www.biketechreview.com
- [3] Andrew S. Gardner, Shaun Stephens, David T. Martin, Evan Lawton, Hamilton Lee, and David Jenkins, Accuracy of SRM and Power Tap Power Monitoring Systems for Bicycling. Med Sci Sports Exerc. 36:1252-1258, 2004
- [4] M.J.A.J.M. Hoes, R.A. Binkhorst, A.E.M.C. Smeeke-Kuyl and A.C.A Vissers, Measurement of Forces Exerted on Pedal and Crank During Work on a Bicycle Ergometer at Different Loads. Internationale Zeitschrift für angewandte Physiologie einschließlich Arbeitsphysiologie Volume 26, Issue 1, pp 33-42. 1968
- [5] Ishida Technologies; Introduction to load Cells. www.ishida.com.
- [6] Picture courtesy of RossoCarbon srl, Italy
- [7]-[10] Pictures courtesy of Breakaway Innovations plc, Perth WA.
- [11] Statement of Accuracy, Verve Cycling Infocrank. Accuwiegh 19 Yampi Way Willerton WA 6155. www.vervecycling.com (Attached as appendix)
- [12] Greg Kopecky Powermeter 301. June 18 2013. http://www.slowtwitch.com/Tech/Power_Meter_301_3696.html
- [13] Picture courtesy of Tough Torque Ltd, Hong Kong.

Appendix.



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Statement of Accuracy

Client: Breakaway Innovations (BAI)
Crank Model: Verve Cycling Infocrank (fitted with Breakaway Innovations KIP module)
Crank Length: 172.5mm
Serial Number: 715
KIP Model: KIP1
HW Revision: 2
SW Revision: 1
Test Weights Used: Orange Set

Load [kg]	Error Summary										
	0	5	10	20	50	100	50	20	10	5	0
Load [Nm]	0.0	8.5	16.9	33.8	84.6	169.2	84.6	33.8	16.9	8.5	0.0
Test	Nm	Nm	Nm	%	%	%	%	%	Nm	Nm	Nm
1	0.00	0.01	0.05	0.19	0.43	0.57	0.17	0.00	0.05	0.01	0.00
2	0.00	0.01	0.02	0.37	0.28	0.48	0.17	0.18	0.08	0.01	0.00
3	0.00	0.04	0.02	0.28	0.39	0.28	0.13	0.19	0.05	0.04	0.00
4	0.00	0.02	0.11	0.37	0.36	0.15	0.20	0.28	0.02	0.01	0.00
5	0.00	0.09	0.05	0.10	0.17	0.02	0.02	0.18	0.02	0.04	0.00
6	0.00	0.06	0.08	0.19	0.28	0.19	0.20	0.19	0.05	0.01	0.00
7	0.00	0.02	0.02	0.10	0.24	0.15	0.13	0.28	0.08	0.01	0.00
8	0.00	0.06	0.08	0.00	0.02	0.10	0.02	0.19	0.05	0.04	0.00
9	0.00	0.06	0.05	0.19	0.28	0.15	0.13	0.18	0.05	0.01	0.00
10	0.00	0.02	0.02	0.10	0.17	0.19	0.20	0.18	0.05	0.04	0.00
Max Errors	0.00	0.09	0.11	0.37	0.43	0.57	0.20	0.28	0.08	0.04	0.00

Absolute Max Error

Torque < 17Nm: 0.11Nm

Torque > 17Nm: 0.57%

A single KIP power meter crank arm was statically tested using standard masses.

A customer supplied crank mounting device was used to hold the cranks horizontally and apply the loads 55mm outboard from the crank face.

The supplied crank was tested and found to have a maximum error of 0.11 Nm below 17 Nm and 0.57% above 17 Nm.

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